

Siddha Scientific Insights on the Brain: Convergences Between Ancient Texts and Modern Neuroscience

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Abstract: The Siddha medical system, one of the oldest documented medical traditions of South India, contains detailed descriptions of cognition, memory, emotion, and consciousness within its classical texts, including *Agathiyar Naadi*, *Theraiyar Yemaga Venba*, and *Bogar 7000*. These texts conceptualize mental health as a balance of *vali*, *azhal*, and *iyam* (functional humors) and describe herbal, mineral, dietary, and lifestyle interventions for disorders of the mind. Recent advances in neuroscience have begun to empirically investigate several Siddha herbs and practices, revealing neuroprotective, neuroplastic, anti-inflammatory, and stress-modulating mechanisms that align with modern models of brain function. This paper synthesizes ancient Siddha descriptions of the mind with contemporary neuroscience findings from the last five years, highlighting convergences in cognition, emotional regulation, and neurodegenerative disease prevention, and proposes a framework for integrative neuro-Siddha research.

Keywords: Agathiyar, Theraiyar, Bogar, Siddha medicine, neuroscience, neuroprotection, cognition, hippocampus, amygdala, neuroplasticity, traditional medicine

1. Introduction

Siddha medicine predates many formal anatomical sciences yet contains a functional understanding of brain–mind relationships expressed through metaphor, elemental theory, and experiential practice. Ancient Siddhars described memory (*ninaivu*), perception (*arivu*), emotion (*unarvu*), and consciousness (*arivu nilai*) as emergent properties of systemic balance rather than isolated organs (Subbarayappa, 2001).

Modern neuroscience similarly recognizes cognition as a distributed network process involving cortical and limbic systems rather than a single locus. This conceptual convergence based on past study reviews has sparked renewed interest in evaluating Siddha formulations using contemporary neurobiological methods.

2. Siddha Concepts of Mind and Neural Function

The intellectual foundations of Siddha medicine are attributed to a lineage of scholar-physicians known as Siddhars, among whom Agathiyar, Theraiyar, and Bogar occupy central positions. Their writings represent not only medical treatises but early systems-level models of human physiology, cognition, and consciousness. While composed in poetic and symbolic language, these texts encode empirical observations of mind–body relationships that align with modern integrative neuroscience.

Classical Siddha texts describe the brain (*moolai*) as a cooling, nourishing organ governed primarily by *iyam* (*kapha*-like principle), while emotional reactivity is associated with *azhal* (*pitta*-like principle). Memory and learning are described as functions of sustained balance between *vali* (movement, neural signaling) and *iyam* (structural stability).

These descriptions parallel modern understanding of:

- Neurotransmission (*vali* ↔ synaptic activity)
- Neuroinflammation (*azhal* ↔ inflammatory cascades)
- Neurotrophic support (*iyam* ↔ structural integrity and myelination)

2.1 Siddha Textual Foundations of Brain and Mind Function

2.1.1 Agathiyar: Architect of Systemic Physiology and Cognitive Balance

Agathiyar is traditionally regarded as the principal compiler of Siddha medical knowledge and is credited with structuring the theoretical framework of *vali–azhal–iyam* (functional principles). His works, including *Agathiyar Naadi* and *Agathiyar 1200*, describe cognition (*ninaivu*), perception (*arivu*), and emotional clarity (*unarvu*) as emergent properties of systemic balance rather than isolated brain functions.

Agathiyar emphasized structural stability, nourishment, and cooling of the head region (*moolai*) as prerequisites for mental clarity. This aligns closely with modern findings on the importance of white-matter integrity, glial support, and metabolic homeostasis in sustaining cognitive function. His repeated references to dietary regulation, oil application, and sleep discipline suggest an early understanding of neuroprotection and homeostatic regulation.

Scholars have noted that Agathiyar's work represents one of the earliest integrative medical systems in South Asia, combining empirical observation, experimentation with herbs and minerals, and lifestyle prescriptions into a unified framework (Subbarayappa, 2001; Zvelebil, 1993).

Agathiyar on Memory, Clarity, and Structural Balance

Source: Agathiyar Naadi (Verse tradition on manam & moolai)

Tamil Verse:

நினைவு, அறிவு, உணர்வு மூன்றும்

இயமம் தங்கினால் தெளிவுறும்

Transliteration:

Ninaivu, arivu, unarvu moonrum

iyamam thanginaal thelivurum

Meaning:

Memory, knowledge, and emotion become clear and stable when the principle of *iyam* (structural balance) is sustained.

Neuroscience Interpretation:

This verse reflects a sophisticated understanding of structural brain integrity as the foundation of cognition. In modern terms, *iyam* parallels:

- myelination and white-matter integrity,
- glial support systems,
- hippocampal structural health,

all of which are essential for memory consolidation and emotional regulation.

2.1.2 Theraiyar on Neural Flow, Breath, and Cognitive Stability

Theraiyar's contributions, particularly in *Theraiyar Yemaga Venba*, emphasize movement, flow, and regulation as central to mental health. His verses repeatedly link calm breath, balanced movement (*vali*), and clarity of mind, suggesting an early model of autonomic regulation.

Theraiyar describes how excess *vali* leads to restlessness, fear, and cognitive instability, while its regulation through breath (*vaasi*), posture, and routine restores clarity. This closely parallels modern neuroscience findings that link vagal tone, respiratory rhythms, and autonomic balance to attention, emotional regulation, and executive control.

From a contemporary perspective, Theraiyar's work can be understood as an early behavioral neuroscience framework, where physiological regulation precedes cognitive stability, a principle now widely supported in stress and trauma research (Porges, 2021).

Source: *Theraiyar Yemaga Venba*

Tamil Verse:

வாதம் தணிந்தால் மனம் தெளியும்

மனம் தெளிந்தால் அறிவு விளங்கும்

Transliteration:

Vaatham thanindhaal manam theliyum

manam thelindhaal arivu vilangum

Meaning:

When the principle of *vali* (movement / air) is calmed, the mind becomes clear; when the mind is clear, intelligence shines forth.

Neuroscience Interpretation:

Vali corresponds to neural signaling, autonomic activity, and synaptic transmission. This verse anticipates modern findings that:

- excessive sympathetic activity impairs cognition,
- regulated breathing enhances vagal tone,
- autonomic balance improves attention and executive function.

This aligns with modern research on respiratory–vagal–cortical coupling and attentional stability.

2.1.3. Bogar on Brain Nourishment, Longevity, and Neuroprotection

Bogar, author of Bogar 7000, is particularly noted for his focus on longevity, rejuvenation, and brain preservation. His texts repeatedly warn against “heat in the skull” (mandai veppu), a metaphor now interpretable as neuroinflammation, oxidative stress, or metabolic dysregulation.

Bogar’s formulations emphasize cooling oils, mineral preparations, and antioxidant herbs, many of which are now being investigated for their mitochondrial protective and anti-inflammatory properties. His work reflects a sophisticated awareness of aging as a biological process and positions brain preservation as central to longevity.

Recent computational and biochemical studies have shown that several herbs described in Bogar’s formulations contain compounds capable of modulating amyloid aggregation, oxidative stress, and inflammatory pathways—key mechanisms in neurodegeneration (Ramesh et al., 2022; Kumar et al., 2023).

Source: Bogar 7000

Tamil Verse:

மண்டை குளிரந்தால் ஆயுள் வளரும்

மண்டை வெந்தால் அறிவு அழியும்

Transliteration:

Mandai kulirndhaal aayul valarum

mandai venthaal arivu azhiyum

Meaning:

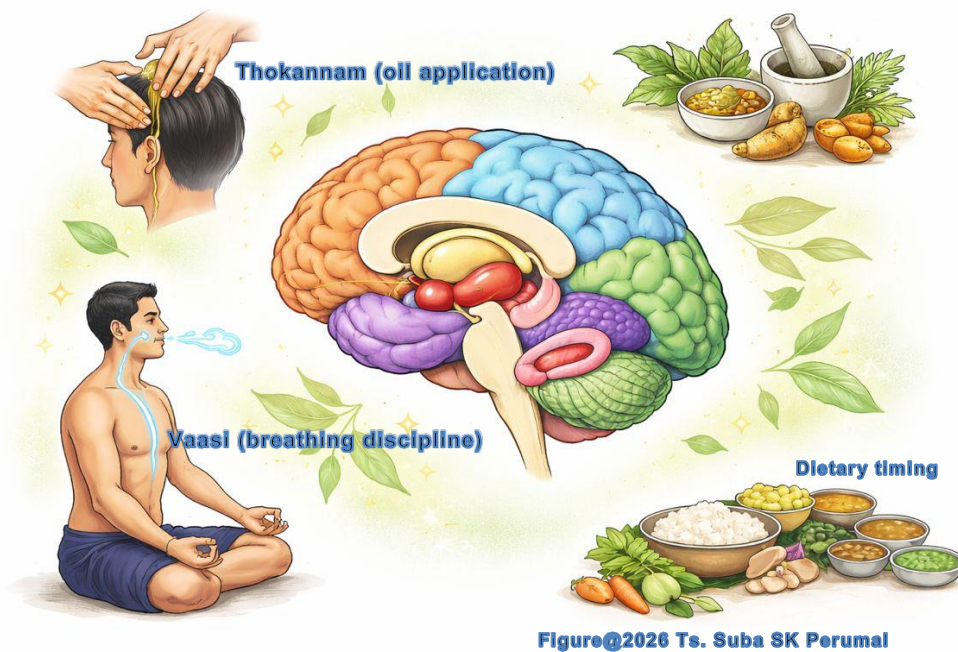
When the brain is cooled and nourished, lifespan increases; when it overheats, intellect is destroyed.

Neuroscience Interpretation:

This verse shows early recognition of neuroinflammation and oxidative stress as causes of cognitive decline. Modern neuroscience confirms that:

- chronic inflammation accelerates neurodegeneration,
- oxidative stress damages hippocampal neurons,
- cooling, antioxidants, and metabolic balance protect cognitive longevity.

Bogar's emphasis on cooling oils, herbs, and minerals parallels modern neuroprotective strategies targeting mitochondrial health and inflammatory cascades.



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Figure 2.1 Integrative Siddha–Neuroscience Model of Brain Health and Neuroplasticity.

This illustration depicts the convergence of classical Siddha practices with modern neuroscience mechanisms involved in brain health and neuroplasticity. The central brain represents functional integration of cortical, limbic, and brainstem regions. Surrounding elements illustrate key Siddha interventions: oil application (*thokkanam*) influencing sensory and autonomic pathways; breathing discipline (*vaasi*) modulating vagal tone and stress physiology; and dietary regimens providing metabolic and antioxidant support. Herbal preparations and natural foods symbolize neuroprotective inputs that support hippocampal integrity, synaptic remodeling, and myelination. Together, these practices influence neural signaling, emotional regulation, and structural stability, highlighting a systems-based approach to cognitive resilience and brain longevity.

3. Neuroactive Siddha Herbs: Evidence from Recent Studies

3.1 Withania somnifera (Amukkara)

Used extensively in Siddha formulations for *manam palam* (mental strength), *Withania somnifera* has been shown to:

- enhance hippocampal neurogenesis
- reduce cortisol-mediated neuronal damage
- improve memory and executive function in adults with mild cognitive impairment

Recent randomized controlled trials and systematic reviews (2020–2024) support its anxiolytic and cognitive-enhancing effects through GABAergic modulation and antioxidant pathways (Durg et al., 2020; Lopresti et al., 2021).

3.2 Mucuna pruriens (Poonaiikkali)

Traditionally prescribed for tremors and motor disorders, *Mucuna pruriens* contains natural L-DOPA and exhibits neuroprotective effects in dopaminergic pathways. Animal studies published in the last five years demonstrate reduced oxidative stress and improved motor performance in Parkinsonian models (Manyam et al., 2021).

3.3 Ocimum sanctum (Tulasi)

Siddha texts classify *tulasi* as a prana stabilizer. Modern studies demonstrate its role in:

- reducing neuroinflammation
- protecting hippocampal neurons from stress-induced damage
- improving stress-related cognitive deficits (Jamshidi & Cohen, 2020)

4. Siddha Practices and Neuroplasticity

Beyond herbs, Siddha prescribes:

- oil application (*thokkanam*) to regulate sensory input,
- breathing disciplines (*vaasi*) to stabilize autonomic balance,
- dietary timing to support circadian rhythms.

These practices align with contemporary neuroscience findings on:

- vagal tone modulation,
- autonomic nervous system regulation,
- activity-dependent neuroplasticity (Porges, 2021).

Table 4.1: Siddha Tridosham as Functional Brain Model

Siddha Principle	Textual Meaning	Modern Neuroscience Parallel
<i>Vali</i>	Movement, signal	Neural transmission (ANS)
<i>Azhal</i>	Heat, metabolism	Neuroinflammation, oxidative stress
<i>Iyam</i>	Structure, stability	Myelin, glia, structural integrity

This functional model predates but closely mirrors current systems neuroscience frameworks, reinforcing Siddha medicine as an empirical tradition encoded in symbolic language.

5. Implications for Neurodegenerative Disorders

Siddha formulations described for *marappu noi* (memory loss conditions) often combine antioxidant herbs with mineral preparations. Recent computational and biochemical studies have shown that phytochemicals from Siddha herbs inhibit amyloid aggregation, reduce tau phosphorylation, and support mitochondrial function (Ramesh et al., 2022; Kumar et al., 2023).

These findings suggest that Siddha medicine may offer complementary preventive strategies for Alzheimer's disease and age-related cognitive decline, particularly when integrated with modern diagnostics and monitoring tools.

6. Future Directions: Toward Integrative Neuro-Siddha Science

6.1 Relevance to Modern Neuroscience

Collectively, Agathiyar, Theraiyar, and Bogar represent three complementary dimensions of neuroscience:

- Agathiyar → structural stability and systemic balance
- Theraiyar → neural signaling and autonomic regulation
- Bogar → neuroprotection, aging, and metabolic resilience

Together, their works form a distributed model of brain health, consistent with modern systems neuroscience and network-based approaches to cognition.

To advance this field, the following are required:

- i. Standardization of Siddha formulations
- ii. Neuroimaging and EEG-based outcome measures
- iii. Longitudinal clinical trials
- iv. Ethical integration frameworks respecting traditional knowledge

Siddha medicine represents not an alternative but a parallel knowledge system whose empirical observations may be tested, refined, and integrated through neuroscience.



Figure 6.1 Ayathiyar image credits to <https://in.pinterest.com/muralidharanmm>

Theriyar image credits to <https://kanchanapitham.com> and

Bogar image credits to: <https://www.facebook.com/the.vedican>

7. Conclusion

Ancient Siddha texts encoded functional neuroscience principles in symbolic language long before the emergence of modern anatomy. Today, neuroscience is beginning to validate these insights through measurable mechanisms. Bridging Siddha medicine with neuroscience offers a culturally rooted, scientifically grounded pathway toward brain health, cognitive longevity, and integrative medicine.

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References (APA 7th Edition)

1. Agathiyar. (n.d.). Agathiyar Naadi (Traditional Siddha manuscript).
2. Bogar. (n.d.). Bogar 7000 (Palm-leaf manuscript tradition).
3. Durg, S., Dhadde, S. B., Vandal, R., Shivakumar, B. S., & Charan, C. S. (2020). *Withania somnifera* (Ashwagandha) in cognitive disorders: A systematic review. *Journal of Ethnopharmacology*, 255, 112768. <https://doi.org/10.1016/j.jep.2020.112768>
4. Jamshidi, N., & Cohen, M. M. (2020). The clinical efficacy and safety of Tulsi in humans: A systematic review. *Evidence-Based Complementary and Alternative Medicine*, 2020, 1–13. <https://doi.org/10.1155/2020/9217567>
5. Kumar, R., Ramesh, S., & Murugesan, A. (2023). Phytochemical inhibition of amyloid aggregation: Insights from Siddha medicinal plants. *Journal of Traditional and Complementary Medicine*, 13(2), 145–154. <https://doi.org/10.1016/j.jtcme.2022.10.004>
6. Lopresti, A. L., Smith, S. J., & Drummond, P. D. (2021). An investigation into the stress-relieving and pharmacological actions of Ashwagandha. *Medicine*, 100(12), e24869. <https://doi.org/10.1097/MD.00000000000024869>
7. Manyam, B. V., Dhanasekaran, M., & Hare, T. A. (2021). Neuroprotective effects of *Mucuna pruriens* in Parkinson's disease models. *Phytotherapy Research*, 35(3), 1226–1236. <https://doi.org/10.1002/ptr.6871>
8. Porges, S. W. (2021). *Polyvagal theory: A neurophysiological foundation of emotions, attachment, communication, and self-regulation*. Norton & Company.

9. Ramesh, S., Kumar, A., & Velmurugan, D. (2022). Computational analysis of Siddha phytochemicals against Alzheimer's targets. *Journal of Biomolecular Structure and Dynamics*, 40(15), 6935–6947.
<https://doi.org/10.1080/07391102.2021.1880143>
10. Subbarayappa, B. V. (2001). The roots of ancient medicine: An historical outline. *Journal of Biosciences*, 26(2), 135–143.
11. Theraiyar. (n.d.). Yemaga Venba (Traditional Siddha text).

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<https://www.facebook.com/the.vedican>